21-242 Recitation 12

RREF — Q8. Bases of (T A) and (T A) from (A)

Course-level solution, with a later-course alternative when useful

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Prompt

Given RREF(A), describe how to find a basis for im(T_A) and ker(T_A).

Course-level solution

Let $R = \text{RREF}(A)$.

- **Image / column space of A.** The pivot columns of R mark the pivot columns of A. Those columns of the *original* matrix A form a basis of im(T_A). (Do not take columns of R: Q4 shows they may span the wrong space.)
- **Kernel / nullspace of A.** Free variables correspond to non-pivot columns. For each free variable x_j, set $x_j = 1$ and set the other free variables to 0; solve the pivot equations (which are immediate from R) for the pivot variables. The resulting n special solutions form a basis of ker(T_A). Equivalently, if the non-pivot columns of R are $R\vec{e}_{j_k}$, each special solution is $\vec{e}_{j_k}$ minus the linear combination of pivot columns that R uses to express that column.

Illustration with the matrix of Q1. $R = \begin{pmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ has pivots in columns $1, 2$. A basis of im(T_A) is the first two columns of A: $(0, 3, 6)^T$ and $(1, 0, 5)^T$. One free variable x_3, and $x_1 + (1/3)x_3 = 0$, $x_2 = 0$, so a basis of the kernel is $\big((-1/3),\, 0,\, 1\big)^T$.